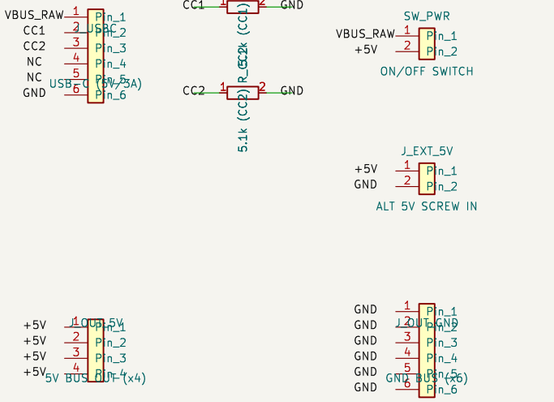
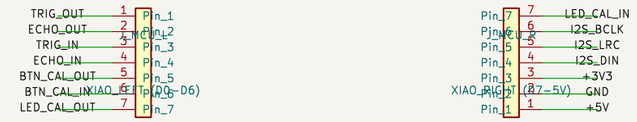
DANEBEN! SMART WASTE TELEMATICS — MODULAR CARRIER BOARD v2.0

Hochschule Niederrhein (HSNR) · 5. Semester · Mikrocontroller · Master Electrical Schematic

DEDICATED USB-C POWER ENTRY & DISTRIBUTION



BLOCK 2: SEEED STUDIO XIAO ESP32-S3 SENSE MCU

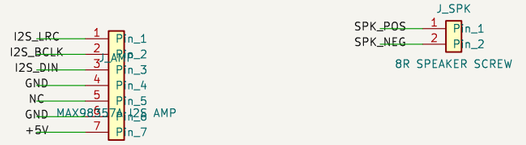


Dual-Core 240MHz ESP32-S3 | 8MB PSRAM

Top-mounted OV2640 AI Camera Subsystem

Native USB-CDC for Telemetry & Flashing

BLOCK 3: I2S AUDIO AMPLIFIER & SPEAKER

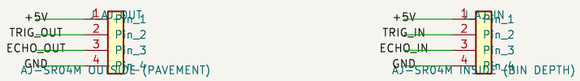


3.2W Class-D Mono DAC/Amp

Direct Digital I2S Audio Bus (No DAC jitter)

2x M2.5 Rigid Standoff Mounts

L AJ-SR04M SENSOR DOCKS (ZERO CABLE FRICTION)

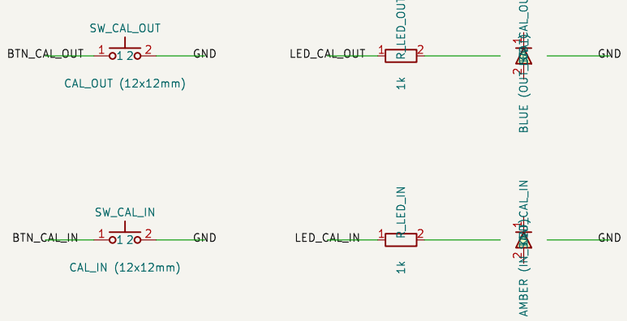


Each dock features 4x M2.5 mechanical standoffs.

Transducer jacks face left PCB edge directly into free air,

guaranteeing ZERO cable bending friction or crowding.

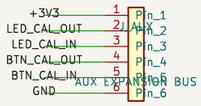
BLOCK 5: SIZABLE CALIBRATION CONTROLS



SW_CAL_OUT: 15-ping pavement distance reference calibration.

SW_CAL_IN: Dynamic container depth calibration (1m to 4m).

BLOCK 6: AUX EXPANSION & CHASSIS



4x M3 Corner Mounting Holes

Board Size: 130.0 mm x 90.0 mm

FR4 2-Layer 1.6mm / 1oz Copper

Duales AJ-SR04M Docking, USB-C Power Entry, 12x12mm Kalibrierungstaster, I2S Audio
Modulares Traegerboard fuer Seeed Studio XIAO ESP32-S3 Sense

Hochschule Niederrhein (HSNR) – Fachbereich Elektrotechnik & Informatik

Sheet: /

File: Daneben_Prototype_Carrier_Board.kicad_sch

Title: Daneben! Prototype Carrier Board (Modular Showcase Edition v2.0)

Size: A3	Date: 2026-10-26	**Rev: v2.0**
KiCad E.D.A. 9.0.4		Id: 1/1